Code No: 157EH **R18**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, July/August - 2022
BASIC MECHANICAL ENGINEERING
(Common to CE, EEE, ECE, CSE, IT, MIE, PTM)

Time: 3 Hours **Max.Marks:75**

Answer any five questions
All questions carry equal marks
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JNTUH Used Papers

1.a) Discuss the first law of thermodynamics applied for a process.
b) "No engine can be made to work on Carnot cycle". Justify the statement.
c) How does transient heat transfer differ from steady heat transfer? How does one-dimensional heat transfer differ from two-dimensional heat transfer? [5+5+5]

2.a) How Natural convection is different from forced convection?
b) State and explain second law of thermodynamics.
c) Write a short on thermodynamic work. [5+5+5]

3.a) Why compression ratio of petrol engine is low compared to diesel engines?
b) What are the desirable characteristics of a good fuel?
c) What will happen, when diesel fuel is accidentally filled to a petrol car? [5+5+5]

4.a) What are the measures of IC engine performance?
b) What are the two types of refrigeration systems? How they differ between each other in terms of the working principles? [5+10]

5.a) Why are hydraulic systems preferred for heavy work than the pneumatic systems?
b) List the fields of applications where fluid power can be used more effectively than any power sources. Explain in short. [7+8]

6.a) Explain the Gear terminology with neat sketch.
b) Write a short note on Gear trains. [10+5]

7.a) Sketch and define transmission angle of a four-bar mechanism.
b) Define the following:
i) Kinematic chain
ii) Mechanism
iii) Structure
iv) Inversion
v) Degree of freedom. [7+8]

8.a) Define the following terms: i) Amplitude ii) Resonance
b) Differentiate between free vibrations and forced vibrations.
c) What are the main causes of vibration? Explain. [5+5+5]

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